

Assessment Test –1

Embedded System (B9)

Date: 21-7-2026

Time : 2 hrs

1. **Write a C program to calculate the average of three integer marks and display the result as a floating-point number.**

Note: Use explicit type casting to obtain the correct average.

Sample Input

Enter three marks: 75 80 90

Sample Output

Average = 81.67

2. **Write a C program to toggle the nth bit of a given integer using bitwise operators.**

Sample Input

Enter the number: 10

Enter the bit position (0-31): 1

Sample Output

Original Number : 10

Modified Number : 8

3. **Find the largest of three numbers using if-else.**
4. **Write a C program to check whether a given integer is Even or Odd using the ternary operator.**

Sample Input

Enter a number: 27

Sample Output

27 is Odd

5. Register Programming

A hardware register is

STATUS = 0xA6

Bit7 : Interrupt

Bit6 : DMA Busy

Bit5 : TX Ready

Bit4 : RX Ready

Bit3 : Error

Bit2 : Overflow

Bit1 : Reserved

Bit0 : Power

Answer the following

- a) Is DMA Busy active? _____
- b) Is RX Ready active? _____
- c) Set Power bit. _____
- d) Clear Error bit. _____
- e) Toggle Overflow bit. _____

Part B:

5*2=10Marks

1. Explain the role of GPIO in an embedded system. Differentiate between input and output GPIO.
2. Differentiate Microprocessor and Microcontroller
3. Define Embedded System draw the basic architecture of an embedded system.
4. Explain the role of CPU, RAM, ROM/Flash, and peripherals in an embedded system.
5. Draw the Embedded System Development Life Cycle.

Part C

(10 × 1 = 10 Marks)

1. Which declaration is valid?

- A) int 2value; B) int _value; C) int float; D) int value-name;

2. Which keyword tells the compiler that a variable may change unexpectedly?

- A) static B) const C) volatile D) register

3. What is the output?

int x = 5;

printf("%d", x++);

- A) 6 B) 5 C) Compilation Error D) Undefined

4. What is the output?

printf("%d", 10/3);

- A) 3.33 B) 3 C) 4 D) Error

5. Which statement immediately exits a loop?

- A) continue B) break C) return D) goto

6. What is the result? 5 & 3

- A) 1 B) 7 C) 2 D) 6

7. Which operator has the highest precedence?

- A) * B) + C) == D) &&

8. unsigned char value=0x5A;

printf("%X", value^0xFF); o/p=_____

9. What is the output of the following program?

```
#include <stdio.h>
```

```
int main()
{
    int x = 4, y = 6, z = 2;

    printf("%d", x > y || y > z && x < z);

    return 0;
}
```

A) 0

B) 1

C) Compilation Error

D) Undefined Behaviour

10. What is the output of the following program? _____

```
#include <stdio.h>
```

```
int main()
{
    volatile int flag = 0;

    int count = 0;

    while(flag == 0)
    {
        count++;

        if(count == 5)
            flag = 1;
    }

    printf("%d", count);

    return 0;
}
```

